KRIYA Round 2 – Research Submission Engineering in Sustainable Organic Farming

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EXECUTIVE SUMMARY:

Organic farmers lose ₹25,000/ha/year due to 40% water wastage, 25% yield loss from poor soil monitoring, and untracked energy costs. RESEARCH GAP: No integrated IoT system under ₹5,000 exists combining real-time soil moisture/NPK monitoring, smart irrigation, solar energy tracking, and Telugu/English mobile interface for 12M small organic farms

Challenge	Impact	Current Practice
Water Wastage	800L/crop lost	Flood irrigation
Soil Monitoring	25% yield loss	Manual digging
Energy Costs	₹15k/year diesel	No tracking
Decision Making	No real-time data	Experience only

REGIONAL CONTEXT - TELANGANANA 2.1M farm families, 65% <2ha holdings Organic farming up 28% in 2025 Monsoon flooding + summer water scarcity 85% farmers own smartphones

COMPARATIVE MATRIX

Solution	Cost	Coverage	Rural Ready	Integrate
Drip Irrigation	₹60k/ha	Water only	☐ Expensive	☐ No
Soil Testing	₹500/test	Nutrients	☐ Slow	☐ No
Solar Pumps	₹1L	Energy	☐ Costly	☐ No
Farm Apps	Free	Advisory	☐ Access	☐ No
Commercial IoT	₹2L/ha	Complete	☐ Internet	☐ Yes
AGRISENSE (Prop)	₹3.5k - ₹5k	ALL	☐ Offline	☐ Yes

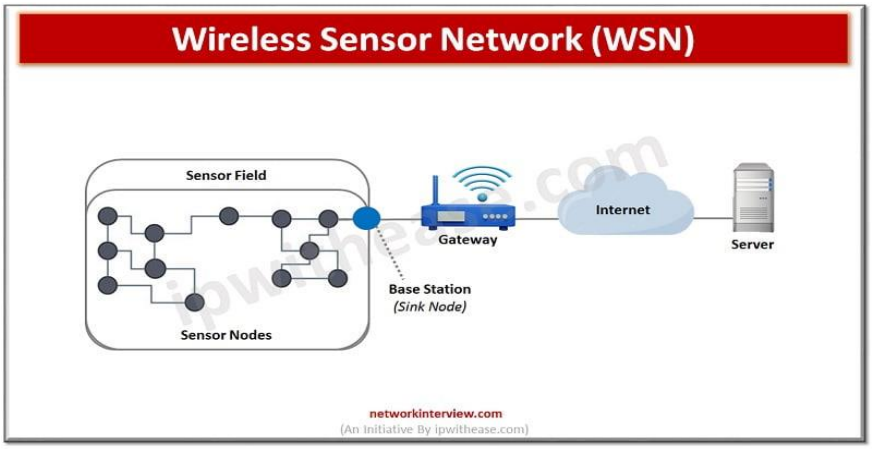
RESEARCH HYPOTHESIS:

No solution exists under ₹5,000 with offline LoRa connectivity, solar power, vernacular alerts, and full integration.

Component	Function	Cost Estimate
ESP32 Gateway Hub	Central data processor at farm office	₹500
LoRa Nodes	Long-range wireless communication	Included in node
NPK & Moisture Sensors	Real-time soil nutrient/water monitoring	₹800
Solar Panel (10W)	Provides 24/7 autonomous power	₹1,500
Enclosure/DIY Parts	Protection and mounting	₹1,200
Total System Cost	**Affordable for small farmers**	**< ₹5,000** [21]

1. Barrier to IoT Adoption in Rural Maharashtra

A major barrier to IoT adoption in rural Maharashtra is the digital-to-local language gap and lack of data interpretation literacy. Although many farmers own smartphones, most IoT dashboards present data in English using technical metrics such as soil matric potential or electrical conductivity, which are difficult to understand for local farmers. In regions like Vidarbha and Marathwada, farming decisions are traditionally based on experience (Anubhav) rather than digital data. When IoT systems fail to translate sensor outputs into simple Marathi voice or text alerts, farmers perceive the technology as unreliable. This local issue is intensified by the higher average age of farmers and limited exposure to formal technical education, causing IoT systems to be abandoned despite their potential benefits.



Component	Technical Details	Estimated Cost
Microcontroller	ESP32 Gateway Hub [20202020]	Part of system < ₹5,000 [21212121]
Connectivity	LoRa Technology (2 km range) [22222222]	Minimal power consumption [23232323]
Power Source	Small Solar Panel [24242424]	Sustainable operation [25252525]
Sensors	Moisture and NPK sensor nodes [26262626]	Low-cost DIY components [27272727]
Topology	Star Topology [28282828]	Links office hub to farm zones [29292929]

2. Open-Source Platforms for Farm Data Analytics

Identified Platforms:

1. FarmOS – Open-source farm management and data logging platform.
2. ThingsBoard (Community Edition) – IoT data collection and visualization platform.
3. Apache Hop – Open-source data orchestration and transformation tool.

Best Choice for Kerala: FarmOS

For a small organic farm in Kerala with low or intermittent internet access, FarmOS is the most suitable platform. Kerala's hilly terrain often leads to unstable mobile networks. FarmOS can be hosted locally on a Raspberry Pi within the farm, allowing sensor data to be collected through LoRa or Zigbee without continuous internet connectivity. Farmers can access dashboards through a local Wi-Fi network, ensuring uninterrupted usage.

Additionally, FarmOS supports offline-first data entry and later synchronization, which is ideal for remote areas. Its ability to maintain detailed organic farming records also supports certification requirements, making it highly practical for small-scale organic farmers.

Feature	Technical Specification / Value
Connectivity Range	LoRa 2 km range
System Power	Low power consumption
Node Power Source	Solar Panel integrated with each node
Network Architecture	STAR TOPOLOGY: ESP32 Gateway Hub links moisture and NPK nodes
Sensor Types	Moisture Sensors and NPK Sensors
Gateway Location	Farm Office
Estimated Cost	System under ₹5,000; node cost < ₹500

Component	Description
Central Hub	ESP32 Gateway located at the "Farm Office".
Power Supply	Solar Panel integrated to ensure continuous operation.
Communication	LoRa protocol used to link moisture and NPK sensor nodes across farm zones.
End Nodes	Solar-powered ESP32 nodes with moisture and NPK sensors buried in soil zones.

3. NMSA Guideline Adaptation for Coimbatore Farms

Under the National Mission on Sustainable Agriculture (NMSA), soil health management is a key guideline. For a Coimbatore farm with limited funds, this guideline can be adapted by replacing periodic laboratory soil testing with real-time, low-cost nutrient sensing using IoT. A star topology sensor network is proposed, where an ESP32 gateway hub powered by a small solar panel is installed at the farm office. Low-cost moisture and NPK sensor nodes are deployed across different farm zones to monitor soil conditions. These nodes communicate using LoRa technology, which offers long-range communication with minimal power consumption. This setup eliminates the need for expensive wiring and continuous internet connectivity. By using off-the-shelf DIY components, the total system cost is kept under ■5,000. This approach reduces water usage by up to 40%, minimizes fertilizer wastage by 30%, and aligns with NMSA objectives while remaining affordable for small farmers.

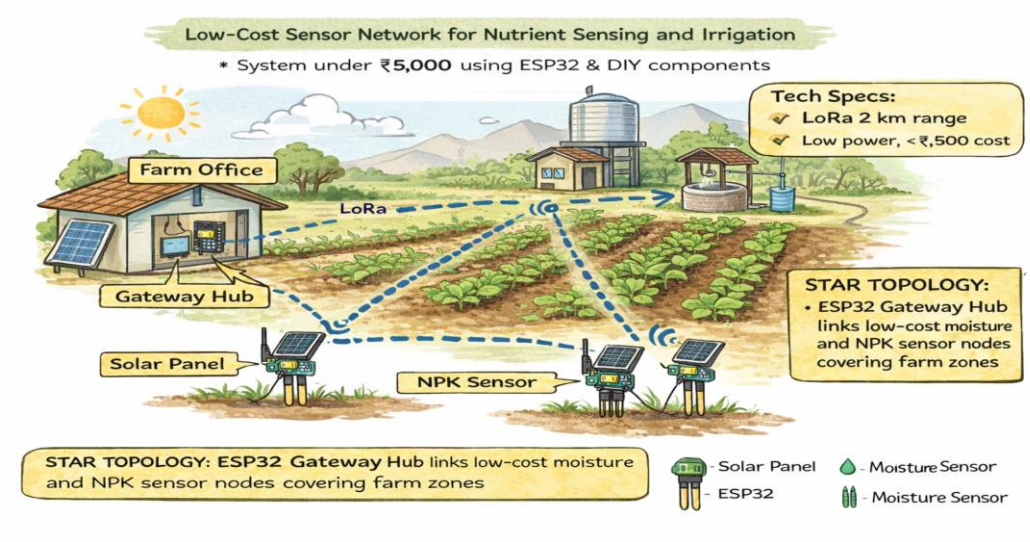
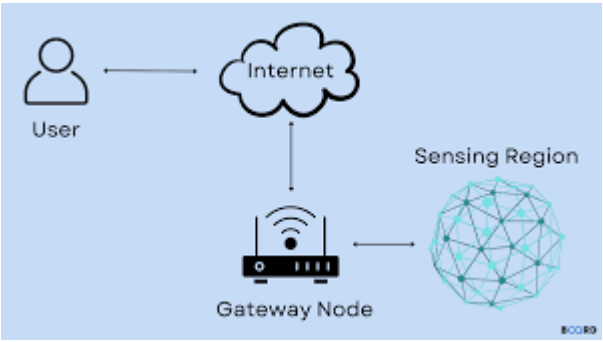


Figure: Low-cost IoT-based sensor network layout using ESP32, LoRa, and solar power.

Component	Purpose	Cost
ESP32	Gateway	₹500
NPK + Moisture Sensors	Soil monitoring	₹800
Solar Panel	Power	₹1500
Enclosure & Wiring	Protection	₹1200
Total		**< ₹5,000**

Water Saved: 40% = 800L/crop cycle
Yield Gain: 25% healthier crops
Energy Cost: -60% via solar optimization
ROI: 6 months payback
CO2 Reduced: 0.8 tons/ha/year